

NS-BLOWUP-F-06 v1: Audit $T_* < \infty \Rightarrow C_F?$

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Mission: critical $\neq$ necessary.

Candidates: $L^3, \dot{H}^{1/2}, BMO^{-1}, B_{\infty,1}^{-1}, B_{\infty,\infty}^{-1}$

Rule: $C_F \in \mathcal{C}_{PROVEN}$ iff $\sup_{[0,T_*)} \|u\|_X < \infty \Rightarrow T_* = \infty$ proven.

Only L^3 via ESS is potential PROVEN. Others OPEN/DEAD.

If none: $\mathcal{C}_{PROVEN} = \{C_E^{weak}\}$ valid. If L^3 : $\mathcal{C}_{PROVEN} = \{C_E^{weak}, C_{L^3}\}$.

No $\perp$ in F-06.